Section 4 — Method: EvidenceFlow（中英对照, v7.4）

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v7.4 主要改进：（1）opening 改写为 **Relink** 风格（method-defining phrase + 一句对比定位 + 三组件 design）；（2）方法定位句：*built around index-grounded evidence propagation*（不喊 paradigm slogan）；（3）对比定位：*Rather than treating multi-hop retrieval as independent passage ranking*（切独立排序路线，不误伤我们 query-time graph construction）；（4）三段保持 §4.1 Offline Indexing / §4.2 Local Graph Retrieval / §4.3 Retrieval Enhancement；（5）**8** 个公式不变；（6）figure ref 暂留 TODO marker，等 overview figure 真画出来再加；（7）ACL 双栏编译 0 overfull。

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§4 Opening

方法节开头

OPENING — RELINK-STYLE: METHOD DEFINITION + CONTRAST + THREE COMPONENTS

We propose **EvidenceFlow**, a graph-based retrieval framework *built around index-grounded evidence propagation* for multi-hop question answering. **Rather than treating multi-hop retrieval as independent passage ranking**, EvidenceFlow grounds entity and relation connectivity in an offline document-grounded OpenIE graph and propagates a query-conditioned *evidence flow* over its query-local view. This design addresses three needs of multi-hop retrieval through three components: (i) offline indexing builds a multi-layer graph that couples passages, entity phrases, OpenIE facts, and provenance (§4.1); (ii) evidence-driven local graph retrieval instantiates a query-local propagation graph and ranks passages by personalized PageRank (§4.2); and (iii) retrieval enhancement combines relation-grounded expansion with coverage-aware reranking to produce the final ranked evidence list Rq (§4.3).	我们提出 **EvidenceFlow**，一个面向多跳问答的 graph-based retrieval 框架，建立在 *index-grounded evidence propagation* 之上。不同于把多跳检索处理为对段落的独立排序，EvidenceFlow 把实体和关系的连接性锚定在离线 **document-grounded OpenIE** 图中，并在该图的 query-local 视图上传播一个 query-conditioned 的 *evidence flow*。这个设计通过三个组件回应多跳检索的三个需求：(i) 离线索引构建一个耦合 passages、entity phrases、OpenIE facts 和 provenance 的多层图（§4.1）；(ii) evidence-driven 的局部图检索实例化一个 **query-local propagation graph**，并以 **personalized PageRank** 给段落打分（§4.2）；(iii) retrieval enhancement 结合 **relation-grounded expansion** 与 **coverage-aware reranking**，产出最终的 ranked evidence list Rq（§4.3）。

§4.1 Offline Indexing

离线索引

LEAD — TWO-STEP OFFLINE CONSTRUCTION

Following standard OpenIE practice (Angeli et al., 2015; Gutiérrez et al., 2025), offline indexing builds a document-grounded graph $\mathscr{G}\mathscr{D}=(\mathscr{V}\mathscr{D},\mathscr{E}\mathscr{D})$ in two steps.	遵循标准 OpenIE 流程（Angeli et al., 2015；Gutiérrez et al., 2025），离线索引分两步构建文档锚定的图 $\mathscr{G}\mathscr{D}=(\mathscr{V}\mathscr{D},\mathscr{E}\mathscr{D})$。

ENTITY AND FACT EXTRACTION

For each document $d \in \mathscr{D}$, a language model extracts entity mentions and OpenIE facts $f=(h,r,t)$, where h and t are head and tail entities and r is the relation phrase connecting them. Entity surface forms are normalized into a phrase-node vocabulary $\mathscr{V}$phr, while every fact retains its provenance src(f)=d. The result is a fact set $\mathscr{F}$ in which each relational assertion remains traceable to the document from which it was extracted.	对每篇文档 $d \in \mathscr{D}$，语言模型抽取实体提及和 **OpenIE facts** $f=(h,r,t)$，其中 h 和 t 是 head/tail 实体，r 是连接它们的关系短语。实体的 surface form 被标准化进 phrase-node 词表 $\mathscr{V}$phr，每个 fact 保留其 provenance src(f)=d。最终得到的 **fact** 集合 $\mathscr{F}$ 中每条 **relational assertion** 都可追溯回它被抽取的源文档。

DOCUMENT-GROUNDED MULTI-LAYER GRAPH

$\mathscr{G}\mathscr{D}$ couples two node layers. *Passage nodes* $\mathscr{V}$doc represent documents in $\mathscr{D}$, and *phrase nodes* $\mathscr{V}$phr represent extracted entities. Three edge types connect them: *passage–phrase* edges attach each passage to the entities it mentions; *relational* edges connect (h,t) for each fact $f=(h,r,t)$, retaining r and src(f) as edge attributes; and *synonymy* edges connect phrase nodes with highly similar dense embeddings. **Passage–phrase edges ground the entity layer in documents, relational edges encode OpenIE assertions among entities, and provenance links each assertion back to the passage that supports it**. We pre-compute dense embeddings for all nodes and materialize $\mathscr{G}\mathscr{D}$ as an adjacency index built once for the corpus; query time performs no further OpenIE extraction over $\mathscr{D}$.	$\mathscr{G}\mathscr{D}$ 耦合两层节点。***Passage nodes*** $\mathscr{V}$doc 表示 $\mathscr{D}$ 中的文档，***phrase nodes*** $\mathscr{V}$phr 表示抽取到的实体。三类边把它们连起来：***passage-phrase*** 边把每篇 passage 连到它提到的实体；关系边对每个 fact $f=(h,r,t)$ 连接 (h,t)，保留 r 和 src(f) 作为边属性；***synonymy*** 边连接 dense 嵌入高度相似的 phrase node。**Passage-phrase** 边把实体层锚定到文档，关系边编码实体间的 **OpenIE** 断言，**provenance** 把每条断言连回支撑它的段落。我们为所有节点预计算 dense 嵌入，并将 $\mathscr{G}\mathscr{D}$ 物化为跨语料一次性建好的邻接索引；query 时不再对 $\mathscr{D}$ 做 OpenIE 抽取。

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§4.2 Evidence-Driven Local Graph Retrieval

证据驱动的局部图检索

LEAD — GQ VS HQ + EVIDENCE FLOW ΠQ

Given q, EvidenceFlow first induces a query-local subgraph Gq from $\mathscr{G}\mathscr{D}$. It then instantiates a local propagation graph Hq over the facts and passages contained in Gq, and ranks passages by personalized PageRank on Hq. We keep these objects separate: **Gq determines which part of the offline passage–phrase graph is considered for the query, while Hq determines the state space on which evidence mass propagates**. The resulting query-conditioned mass distribution πq, which we call the *evidence flow*, is therefore both local to q and grounded in the offline graph.	给定 q，EvidenceFlow 先从 $\mathscr{G}\mathscr{D}$ 诱导出 **query-local** 子图 **Gq**。然后在 Gq 包含的 facts 和 passages 之上实例化 **local propagation graph Hq**，并在 Hq 上跑 personalized PageRank 给 passage 排序。我们故意把这两个对象分开：**Gq** 决定离线 **passage-phrase** 图的哪一部分进入 **query** 考虑范围，**Hq** 决定 **evidence mass** 在哪个状态空间上传播。由此得到的 query-conditioned mass 分布 πq——我们称之为 ***evidence flow***——既 **local** 于 **q**，又锚定在离线图上。

QUERY-LOCAL SUBGRAPH

We extract entity anchors $A(q) \subseteq \mathcal{V}_{phr}$ from q using **a controlled language model that operates on the question only, never on retrieved documents**. A standard dense retriever provides an entry signal $\mathrm{Entry}(q) \subseteq \mathcal{D}$. **The entry signal is used only as an activation prior**: it seeds the local graph but does not by itself determine the final ranking. The subgraph G_q is initialized with phrase nodes corresponding to $A(q)$, passages in $\mathrm{Entry}(q)$, and any passage that is the provenance of a relational edge incident to $A(q)$. From this seed set, G_q expands along relational, passage–phrase, and synonymy edges of $\mathcal{G}_{\mathcal{D}}$ for a bounded number of hops. Thus G_q contains the passage and phrase nodes that can participate in query-time retrieval, while excluding the rest of the corpus graph. This keeps the activated region compact while allowing passages reachable through multi-hop entity chains to enter the local retrieval space even when their surface similarity to q is low.	我们用仅作用于问题、绝不作用于检索文档的受控语言模型从 q 抽取 entity anchors $A(q) \subseteq \mathcal{V}_{phr}$。一个标准 dense retriever 给出 entry signal $\mathrm{Entry}(q) \subseteq \mathcal{D}$。**entry signal** 只作激活先验：它为 local graph 播种但不单独决定最终排序。子图 G_q 初始化为：(i) $A(q)$ 对应的 phrase nodes；(ii) $\mathrm{Entry}(q)$ 中的 passages；(iii) 任何作为某条 incident 到 $A(q)$ 的关系边 provenance 的 passage。从这个种子集出发，G_q 沿 $\mathcal{G}_{\mathcal{D}}$ 的关系边、passage–phrase 边和 synonymy 边做有界跳数扩展。这样 G_q 包含可参与 query-time 检索的 passage 和 phrase 节点，同时排除语料图的其他部分。激活区域紧凑，但仍允许那些只能通过多跳实体链才能到达的 **passage** 进入局部检索空间，即便它们对 **q** 的表面相似度低。

LOCAL PERSONALIZED PAGERANK (HQ + 2 公式)

From Gq, we construct Hq with two node types: local fact nodes and local passage nodes. **A fact f=(h,r,t)∈$\mathcal{F}$ belongs to $\mathcal{F}$q when its provenance src(f) is a passage node in Gq.** Because the passage layer of Gq has already been activated by the question-conditioned construction described above, $\mathscr{F}$q inherits the query-locality of Gq through provenance without requiring a separate phrase-side condition. Two fact nodes are linked when their facts share an entity endpoint, and each fact node links to its provenance passage. This derived graph lets propagation move through facts while still scoring passages as the retrieval output. The seed distribution sq places mass on facts whose arguments include an anchor in A(q), weighted by entry scores of the corresponding passages. We solve, by power iteration on Hq,

> πq=αsq+(1−α)Pq⊤πq,(1)

where Pq is the transition matrix induced by fact–fact and fact–passage links in Hq. The passage score for d aggregates stationary mass on local facts grounded in d:

> sppr(d∣q)=|$\mathscr{F}$q,d|−1/2∑f∈$\mathscr{F}$q,dπq(f).(2)

Here, $\mathscr{F}$q,d={f∈$\mathscr{F}$q:src(f)=d}. The square-root normalization keeps the passage score robust to deduplication of the underlying fact set. Because propagation is restricted to Hq, sppr rewards passages connected to q through entity-relation evidence rather than passages independently similar to q.

从 Gq 出发，我们构造 Hq，含两类节点：local fact 节点和 local passage 节点。一个 **fact f=(h,r,t)**∈$\mathcal{F}$ 属于 $\mathcal{F}$**q** 当且仅当其 **provenance src(f)** 是 **Gq** 中的 **passage** 节点。由于 Gq 的 passage 层已经被上述 question-conditioned 构造激活，$\mathscr{F}$q 通过 **provenance** 继承 **Gq** 的 **query-locality**，不需要单独的 **phrase-side** 条件。两个 fact 节点当其 fact 共享一个实体端点时相连；每个 fact 节点连接到它的 provenance passage。这个派生图让传播在 **facts** 上移动，但仍以 **passages** 作为检索输出来打分。种子分布 sq 把质量放在 argument 包含 A(q) 中锚点的 fact 上，按对应 passage 的 entry score 加权。我们在 Hq 上用 power iteration 求解：

> πq=αsq+(1−α)Pq⊤πq,(1)

其中 Pq 是由 Hq 中 fact–fact 和 fact–passage 链接诱导的传播矩阵。passage d 的分数聚合锚定到 **d** 的 **local facts** 的稳态质量：

> sppr(d∣q)=|$\mathscr{F}$q,d|−1/2∑f∈$\mathscr{F}$q,dπq(f).(2)

其中 $\mathscr{F}$q,d={f∈$\mathscr{F}$q:src(f)=d}。平方根归一化让 **passage** 分数对底层 **fact** 集合的去重保持鲁棒。由于传播被限制在 Hq 上，sppr 奖励通过实体-关系证据连到 **q** 的 **passage**，而不是仅仅对 **q** 独立相似的 **passage**。

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§4.3 Retrieval Enhancement with Relational Evidence

基于关系证据的检索增强

LEAD — TWO FAILURE MODES OF LOCAL-PPR

Local propagation yields an initial graph-aware ranking, but two failure modes remain. First, a passage on a multi-hop chain can receive low πq-mass when it is several hops away from any anchor. Second, the top of the ranking can be dominated by passages that cluster around a single entity neighborhood, leaving other entities or relations demanded by q under-covered. We address both with two enhancements that operate on top of sppr.

局部传播给出了初始的图感知排序，但仍有两种失败模式。第一，多跳链上离 **anchor** 较远的 **passage** 可能获得低 **πq** 质量，即便它携带闭合该链所需的关系证据。第二，排序顶部可能被聚集在单一实体邻域的 **passage** 主导，让 q 所需的其他实体或关系覆盖不足。我们用两个作用在 **sppr** 之上的增强同时处理两种问题。

ENHANCEMENT 1 — RELATION-GROUNDED EXPANSION (3 公式)

For each high-scoring fact in π_q, we collect its incident phrase nodes in G_q and follow nearby relational edges of $\mathcal{G}_\mathcal{D}$. Their provenance passages are admitted as expansion candidates. For a relational edge e between phrase endpoints (h_e,t_e) with relation phrase r_e and provenance $src(e)=d$, let

$$m_q(e)=\max\{\pi_q(f): f=(h,r,t)\in\mathcal{F}_q, \{h,t\}\cap\{h_e,t_e\}\neq\emptyset\}, \quad (3)$$

with $m_q(e)=0$ if no such fact exists. The expansion score combines this mass with the dense compatibility between r_e and q:

$$s_{exp}(d\mid q)=\max_{e:src(e)=d} m_q(e)[\sigma(\mathbf{r}_e,\mathbf{q})]_+, \quad (4)$$

where $\mathbf{r}_e$ and $\mathbf{q}$ are dense embeddings of the relation phrase on edge e and the question, respectively; σ is cosine similarity; and $[x]_+=\max(x,0)$ clips to non-negative values **so that expansion can only add evidence**. We combine local-PPR and expansion evidence into a base score

$$s_{base}(d\mid q)=s_{ppr}(d\mid q)+\eta s_{exp}(d\mid q). \quad (5)$$

This step admits bridge and tail-entity passages required by the relational structure of q even when they are not lexically similar to q. Let C_q denote the union of passages scored by local-PPR and passages admitted by relation-grounded expansion; passages without an expansion path have $s_{exp}(d\mid q)=0$.

对 π_q 中每个高分 fact，我们收集其在 G_q 中的 incident phrase 节点，沿 $\mathcal{G}_\mathcal{D}$ 的邻近关系边追踪。这些边的 provenance passage 作为扩展候选被准入。对一条 phrase 端点 (h_e,t_e)、关系短语 r_e、provenance $src(e)=d$ 的关系边 e，定义

$$m_q(e)=\max\{\pi_q(f): f=(h,r,t)\in\mathcal{F}_q, \{h,t\}\cap\{h_e,t_e\}\neq\emptyset\}, \quad (3)$$

若不存在这样的 fact 则 $m_q(e)=0$。扩展分数把这个质量与 r_e 跟 q 之间的 dense 兼容度结合：

$$s_{exp}(d\mid q)=\max_{e:src(e)=d} m_q(e)[\sigma(\mathbf{r}_e,\mathbf{q})]_+, \quad (4)$$

其中 $\mathbf{r}_e$ 和 $\mathbf{q}$ 分别是边 e 上的关系短语和问题的 dense 嵌入；σ 是 cosine 相似度；$[x]_+=\max(x,0)$ 把负值 clip 掉，这样扩展只能往证据池子里加东西。我们把 local-PPR 和扩展证据合成 base score：

$$s_{base}(d\mid q)=s_{ppr}(d\mid q)+\eta s_{exp}(d\mid q). \quad (5)$$

这一步把那些 $\mathbf{q}$ 关系结构所需要、但跟 $\mathbf{q}$ 词汇不相似的桥 **passage** 和尾实体 **passage** 拉进来。令 C_q 表示 local-PPR 打分的 passages 和 relation-grounded expansion 准入的 passages 的并集；没有扩展路径的 passage 有 $s_{exp}(d\mid q)=0$。

ENHANCEMENT 2 — COVERAGE-AWARE RERANKING (3 公式)

We extract a small set of question-side requirements $B(q)=\{b_1,b_2,\ldots\}$ from q via the same controlled language model used for $A(q)$; each $b\in B(q)$ is an entity or relation requirement that the evidence should help resolve. **This extraction is applied to the question only and never generates evidence chains over retrieved text.** For a passage d and requirement b, a contribution $\phi_{d,b}\in[0,1]$ is derived from dense similarity between the passage and the requirement under q. The coverage of a selected set R for requirement b is computed under a noisy-OR model,

$$\mathrm{cov}_q(R\mid b)=1-\prod_{d\in R}(1-\phi_{d,b}),\quad(6)$$

which saturates at 1 once any selected passage substantially contributes to b. Writing $\delta_{q,b}(d\mid R)=\mathrm{cov}_q(R\cup\{d\}\mid b)-\mathrm{cov}_q(R\mid b)$ for the per-requirement gain, the marginal coverage gain of adding d is

$$\Delta_q(d\mid R)=\sum_{b\in B(q)}\delta_{q,b}(d\mid R).\quad(7)$$

We form R_q greedily, at each step adding the candidate d^* that maximizes

$$d^*=\arg\max_{d\in C_q\setminus R}\lambda s_{\mathrm{base}}(d\mid q)+(1-\lambda)\Delta_q(d\mid R).\quad(8)$$

Because the per-requirement coverage saturates, passages duplicating already-covered neighborhoods receive diminishing marginal gain, while passages addressing previously-uncovered requirements are promoted. The final output is the ranked evidence list R_q over passages in $\mathcal{D}$; downstream systems may consume any prefix of R_q.

我们从 q 抽取一小组 question-side requirements $B(q)=\{b_1,b_2,\cdots\}$，使用跟 $A(q)$ 相同的受控语言模型；每个 $b\in B(q)$ 是证据应该共同解决的实体或关系槽位。这一抽取作用在问题上，绝不在检索到的文本上生成 **evidence chain**。对一个 passage d 和 requirement b，定义贡献 $\phi_{d,b}\in[0,1]$，由 d 跟 b 在 q 下的 dense 相似度导出。一个选中集合 R 对 requirement b 的覆盖率用 noisy-OR 模型计算：

$$\mathrm{cov}_q(R\mid b)=1-\prod_{d\in R}(1-\phi_{d,b}),\quad(6)$$

一旦 R 中有 passage 对 b 贡献足够大，覆盖率饱和到 1。记 $\delta_{q,b}(d\mid R)=\mathrm{cov}_q(R\cup\{d\}\mid b)-\mathrm{cov}_q(R\mid b)$ 为单个 requirement 的增益，则加入 d 的边际覆盖增益为：

$$\Delta_q(d\mid R)=\sum_{b\in B(q)}\delta_{q,b}(d\mid R).\quad(7)$$

我们以贪心方式构造 R_q，每一步加入最大化以下目标的 d^*：

$$d^*=\arg\max_{d\in C_q\setminus R}\lambda s_{\mathrm{base}}(d\mid q)+(1-\lambda)\Delta_q(d\mid R).\quad(8)$$

由于单个 **requirement** 的覆盖率饱和，邻域已被覆盖的 **passage** 获得递减边际增益，而处理先前未覆盖 **requirement** 的 **passage** 被提升。最终输出是对 $\mathcal{D}$ 中 **passage** 的 **ranked evidence list R_q**；下游系统可消费 R_q 的任意前缀。

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